Lextract: LLM-Based Extraction of Relevant Market Definitions from European Commission Decisions with Source-Grounded Quality Control

Shriyan S. YAMALI
Independent Researcher, United States
ORCiD ID: Shriyan S. Yamali https://orcid.org/0009-0002-4861-7264

Abstract. Relevant market definitions are central to merger and antitrust analysis, but collecting them at scale from European Commission decisions is difficult because the decisions are numerous and heterogeneous in structure and wording. We present Lextract, an open-source pipeline that filters decision PDFs, uses Gemini 2.0 Flash in two calls to locate the market-definition section and to isolate individual definitions, and stores the results as JSON. Applied to 600 cases, it produced 4,105 records that feed JurisMercatus, a semantic-search interface. We document prompts, settings, exclusion rules and thresholds, give a worked example from source text to record, and combine the model with symbolic quality control: a structural validity test derived from 50 manually checked definitions (76.6% of records pass), a source-fidelity test that checks whether each record occurs verbatim in its decision, and a manual inspection of 100 records. Checked against all 699 retrievable decisions, 79% of the 4,067 verifiable records are verbatim and 9% near-verbatim; 12% are absent from their source. Three quarters of the absent records come from short-form simplified-procedure decisions that evaded the phrase filter, after which the model fabricated plausible definitions, and 30 are copies of the prompt's own examples. The structural test accepts 71% of the absent records and so cannot detect them; the fidelity test also caught one fabricated record in the manually checked reference set and yields a cleaned release of 3,607 records. We describe the fixes and state what remains unmeasured: a gold-standard precision/recall study and baseline comparisons, for which annotation sheets and scripts are provided.

Keywords. legal information extraction, competition law, relevant market definition, large language models, quality control, European Commission

1. Introduction

Relevant market definition is a basic step in merger and antitrust analysis: it fixes the products and the geographic area within which market power is assessed [1]. Practitioners and researchers therefore need to know how the European Commission has delineated a given market before: whether retail distribution of cars is national or EEA-wide, whether a product market was segmented or "left open", and how such findings changed

over time. This is a concrete user problem: the Commission has published more than 6,000 merger and antitrust decisions [2] and adds roughly 280 a year [3]; the market-definition discussion sits under different headings and in different wording, so keyword search over PDFs is unreliable and manual collection is slow and hard to reproduce. The only dedicated resource we know of, the Caselex Market Definitions Module [4], is a paid, editorially curated web product without bulk or programmatic access.

This paper presents *Lextract*,[1] an open-source pipeline that extracts market-definition passages from Commission decision PDFs with a large language model (LLM) and stores them as structured records, and the 4,105-record collection behind *JurisMercatus*,[2] a semantic-search interface. The design follows the hybrid view that LLMs are good at locating and copying legally relevant passages but that their output must be checked by deterministic means. An earlier version reported only a structural check and was rightly criticised for not measuring whether extracted passages are correct. This version contributes:

- a complete description of the pipeline (prompts, model and settings, exclusion phrases, thresholds) and a worked example from source text to record (Sections 3–4);
- a description of the released collection and of what researchers can do with it (Section 5);
- a three-part evaluation: the structural validity test, a new *source-fidelity* test that checks whether each record occurs verbatim in its decision, and a manual inspection of 100 records (Sections 6–7);
- an error analysis identifying two failure classes invisible to the structural test, fabricated definitions for short-form decisions and leakage of prompt examples, with the fixes adopted (Section 8), and a statement of what remains unmeasured together with the annotation protocol and baseline scripts needed to close the gap (Section 9).

2. Related Work and Existing Resources

Legal information extraction combines deterministic, statistical and, more recently, generative methods. Agnoloni et al. presented the BO-ECLI Parser for extracting and linking legal references from European case law [5]; Darji et al. released annotated German legal-reference data and stressed the role of domain-specific gold data [6]. Recent work moves to semantic structuring with LLMs: Janatian et al. turned legislation into structured pathways for expert systems [7]; Costa et al. extracted elements from Brazilian judicial decisions, evaluated them with precision and recall against expert annotations and published them with provenance [8]; Spada et al. extracted events and dates from EU procurement notices with expert validation [9]; others address data scarcity and context [10,11,12]. Lextract shares their pipeline view but targets complete market-definition passages and emphasises cheap deterministic checks of LLM output, which the hallucination literature motivates [13].

[1] `https://github.com/shriyanyamali/Lextract/tree/lda2026`
[2] `https://jurismercatus.shriyanyamali.com`

Relation to existing resources. Caselex [4] offers human-reviewed market definitions across many authorities and languages behind a subscription and a web interface only. Lextract is not a substitute for that editorial product; it offers a free, reproducible and updatable collection with programmatic access: the pipeline can be re-run on new decisions, every record carries case metadata and the decision URL, and an export tool writes CSV, JSONL, Parquet, SQLite, BibTeX and RIS. To our knowledge no openly licensed, machine-readable collection of Commission market definitions existed before.

3. The Lextract Pipeline

Lextract is a sequence of six Python scripts driven by `run_pipeline.py`. The prompts are reproduced verbatim in the repository (`docs/PROMPTS.md`) and summarised here.

3.1. Document Collection and Filtering

Step 1 reads the Excel export of the Commission's competition case-search portal and extracts, per case, the decision URL, case number, year and policy area (inferred from the URL path). *Step 2* downloads each PDF and extracts its text with PyPDF2. A document is excluded if it has exactly three pages and contains one of five phrases from the closing recital of short-form decisions adopted under the simplified procedure [14]: "For the reasons set out in the Notice on a simplified", "European Commission has decided not to oppose the notified operation", "declare it compatible with the internal market", "This decision is adopted in application of Article 6(1)(b)", and "Merger Regulation and Article 57 of the EEA Agreement". Such clearances contain no market analysis, so any "definition" extracted from them would be spurious. Included documents are written to batch files with a metadata header and split into two groups at 80,000 characters so that long decisions can be processed separately; no other length filter is applied. Section 8 shows that this phrase filter was too brittle; the released version normalises whitespace before matching and applies to documents of at most three pages.

3.2. Section Identification (LLM call 1)

Step 3 sends each batch file to Gemini 2.0 Flash (`gemini-2.0-flash` via the `google-generativeai` SDK with default generation settings: no temperature, top-p or output-length override, no system instruction, no JSON mode). The prompt asks the model to find the first occurrence of the words "market definition", to start one line before it, to continue past sub-headings and to stop at the next heading of the same level (e.g. from "IV. Market Definition" to "V. . . . "), excluding that heading, and to reproduce the metadata header unchanged; the output is the market-definition section as plain text.

3.3. Definition Extraction (LLM call 2)

Step 4 sends each section to the same model with a prompt that asks for a JSON array of objects with the fields `case_number`, `year`, `policy_area`, `link`, `topic` and `text`. It instructs the model to extract "the entirety of each individual market definition" as "a direct quote of the multiple paragraphs. Not a paraphrased or summarized version"; to

Table 1. Worked example from Case M.9324. Top: PyPDF2 text of the decision (footnotes and later paragraphs omitted). Bottom: the record produced by Lextract.

Source text (decision PDF, Section 4)

```
4. PRODUCT MARKET DEFINITION
4.1.1. Wholesale distribution of ICT and CE products
(12) In the precedents listed at paragraphs (13)-(15), the Commission
assessed the possible product markets for wholesale distribution of IT
products, telecommunication equipment and CE.
(13) In Avnet/Magirus,8 Tech Data/Scribona,9 Tech Data/Avnet,10 the
Commission has defined a market for the wholesale distribution of
IT products, but left open the question whether the market should be
further sub -segmented (i) by product categories (e.g. server, storage,
PCs, etc.), (ii) by direct and indirect sales11 or (iii) by different
distribution models (broadline vs value -added distribution ("VAD")).12
(14) In Argues/SHC13 and ALSO/PCF,14 the Commission identified a
separate market [...]
```

Extracted record


```
{"case_number": "M.9324", "year": "2019", "policy_area": "Merger",
 "link": "https://ec.europa.eu/competition/mergers/cases1/202048/m9324_
479_3.pdf",
 "topic": "Wholesale distribution of IT products",
 "text": "In Avnet/Magirus, Tech Data/Scribona , Tech Data/Avnet ,
the Commission has defined a market for the wholesale distribution of
IT products, but left open the question whether the market should be
further sub -segmented (i) by product categories (e.g. server, storage,
PCs, etc.), (ii) by direct and indirect sales or (iii) by different
distribution models (broadline vs value -added distribution ("VAD")).”}
```


treat “left open” passages as definitions; to drop paragraph numbers, footnote markers and footnotes; to re-join words split by PDF extraction; to invent a short `topic`; and it states that a definition “is always at least 4 sentences long, but never longer than 12 sentences”. It contains four worked examples (detergents, chemical intermediates, flame retardants, trimethylolpropane). The two-stage design assumes that the second call performs better on a few pages of relevant text than on a 50-page decision; this has not been tested against a single-call design (Section 9).

3.4. Cleaning and Merging

Step 5 strips Markdown code fences from the model’s output and *Step 6* concatenates all files into `data/output.json`. No de-duplication, truncation or editing of the text is performed, so the released records are exactly what the model returned.

4. Worked Example

Table 1 follows one passage through the pipeline. The source is Case M.9324 *ALSO / ABC Data Group* (12 June 2019), a 16-page Article 6(1)(b) decision. Call 1 located

Table 2. Records per year and structural validity rate.

Year	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	All
Records	9	66	621	492	699	594	691	618	297	18	4,105
Valid	3	45	467	421	551	477	509	472	184	14	3,143
Rate (%)	33	68	75	86	79	80	74	76	62	78	76.6

the heading "4. PRODUCT MARKET DEFINITION" and returned Section 4 up to, but excluding, "5. COMPETITIVE ASSESSMENT". Call 2 returned seven records; the one shown is paragraph (13). The model removed the paragraph number and footnote markers 8–12, kept the wording otherwise intact, including the extraction artefact "sub -segmented", and labelled the topic. The structural test accepts the record (63 words; learned phrase "the commission has"; marker "left open"), and the fidelity test finds it verbatim in the decision (coverage 1.0). The sibling record for paragraph (14) is only *near*-verbatim (coverage 0.83) because in the PyPDF2 text that paragraph is interrupted by a page-break footnote block, which the model correctly skipped; the other five records for this decision are verbatim.

5. The JurisMercatus Collection

The collection (`data/jurismercatus_definitions.json` in the repository) contains 4,105 records from 600 cases (578 merger, 22 antitrust) and 704 distinct decision documents dated 2016–2025 (Table 2). The documents were exported from the case-search portal in English; the exact export query of the original run was not recorded, which we note as a reproducibility gap, but the released links identify the processed documents exactly. Cases contribute 1 to 45 records (median 6); records range from 9 to 1,182 words (median 67, mean 91). The `topic` field holds 3,250 distinct free-text labels. The collection is not de-duplicated: 117 records repeat the text of another record, 89 of them within the same case (one passage emitted under two topic labels). PDF-extraction artefacts such as "in -house" survive in 30% of records. A cleaned release (`data/jurismercatus_definitions_clean.json`; 3,607 records, 455 cases) drops the records that the fidelity test (Section 7) finds absent from their source and all records from documents of at most three pages.

What researchers can do with it. Each record pairs a passage with its case number, year, policy area and decision URL, so it can be cited and checked. Through JurisMercatus, a query such as "retail distribution of new passenger cars, geographic scope" returns the passages in which the Commission discussed that question across cases, ranked by embedding similarity. Exported to a spreadsheet or SQLite database, the records support questions such as how often the Commission leaves a definition open by sector and year, or how geographic findings (national, EEA-wide, worldwide) are distributed.

6. Evaluation Design

6.1. Reference Set and Structural Validity Test

No gold standard listing all market definitions in a set of decisions exists, so we use three deliberately limited instruments and say what each measures. The reference set (`evaluation/reference.json`) holds 50 records that the author selected from the pipeline output and checked against the source PDFs: a convenience sample, not a random one, of 39 cases, all merger decisions, mostly 2022–2024, 33 to 557 words long (median 135). Re-examining it for this paper revealed one duplicated record, two case numbers without the "M." prefix (one mistyped) and, through the fidelity test, one record that does not occur in its source (Section 7).

The structural test (`evaluate.py`) derives three checks from the reference set. *Length*: 26 to 668 words, the reference range widened by 20% on each side. *Legal phrase*: at least one of the 121 word n-grams ($n = 2$–5) that occur in at least 15% of reference records (e.g. "can be left open", "for the purposes of this decision"; many are function-word sequences such as "of the"). *Structural marker*: at least one of 18 fixed markers: "relevant market", "product market", "geographic market", "market definition", "left open", "EEA-wide", "EEA wide", "national market", "Commission considers", "Commission found", "Commission has previously", "Notifying Party", "internal market", "serious doubts", "plausible market", "exact scope", "for the purposes of", "cannot be excluded". Seven of the 50 reference records fail the marker check themselves, so the markers are a strict rather than a lenient proxy. *What it measures:* whether a record resembles a verified market-definition passage in length and phrasing. *What it does not measure:* whether the text occurs in the decision (fabrication), whether its boundaries are right, whether it is the right passage, or whether definitions were missed. We report a *structural validity rate*, not accuracy.

6.2. Source-Fidelity Test

The fidelity test (`verify_fidelity.py`) asks whether a record is copied from its decision. Record and PyPDF2 decision text are normalised to lower-case letters only, which removes whitespace, digits and punctuation and so neutralises paragraph numbers, footnote markers and split-word artefacts. The record is cut into 40-letter windows at stride 20, and *coverage* is the fraction of windows found in the source: *verbatim* (≥ 0.95), *near-verbatim* (0.80–0.95), *partial* (0.50–0.80) or *absent* (< 0.50). The test cannot distinguish a copy from text differing only in digits or punctuation, nor judge whether the copied passage is the *right* one. It was run on every record whose decision could be fetched from the stored URL in October 2026: 699 of the 704 documents and 4,067 of the 4,105 records (four URLs of case M.11241 now redirect and one points to an HTML index page).

6.3. Manual Inspection

To learn what the structural test's verdicts mean, 50 passing and 50 failing records were drawn at random (seeded) and read. Each was labelled from the record text alone, without the decision, as A1: a market-definition statement (precedent, party submission, market-investigation finding or Commission conclusion on product or geographic

Table 3. Source-fidelity test on all retrievable records and on the reference set's source documents (five URLs, 38 records, were unretrievable).

	Collection	Reference set
Documents checked (of which ≤3 pages)	699 (136)	38 (1)
Records checked	4,067	48
Verbatim (coverage ≥0.95)	3,221 (79.2%)	44
of which exact match (1.0)	2,985 (73.4%)	–
Near-verbatim (0.80–0.95)	349 (8.6%)	3
Partial (0.50–0.80)	26 (0.6%)	0
Absent (<0.50)	471 (11.6%)	1
Absent, from ≤3-page documents	353 of 380	1 of 1
Absent, from longer documents	118 of 3,687	0 of 47

scope); A2: reasoning within a market-definition analysis without a definitional claim; B: a fragment such as a bare heading; or C: not market definition (competitive assessment, remedies, party description, legal background). The labels are released with the sample (`evaluation/manual_inspection_sample.csv`). This is a single-annotator face-validity reading, not a gold standard, and borderline cases exist: a conclusion that a transaction raises no serious doubts "regardless of the exact product market definition" was labelled C, a list of "affected markets" A1. Precision, recall and F1 require, for whole decisions, the list of every passage that counts as a definition, and what counts, at which unit (paragraph, sub-section, conclusion only), changes the scores. The repository contains an annotation protocol and a script that produces, for N random decisions, a sheet listing every record next to the decision text with columns for true positive, boundary error, false positive and missed definition (Section 9).

7. Results

7.1. Structural Validity

Of the 4,105 records, 3,143 (76.6%) pass all three checks. The 962 failures decompose as 668 marker only, 185 length only, 80 length and marker, 19 phrase and marker, and 10 all three; of the 275 length failures, 261 are shorter than 26 words and 14 longer than 668. The yearly rate (Table 2) lies between 68% and 86% in every year with more than 50 records except 2024 (62%); the 65 antitrust records also reach only 62%. Among the 777 records without a marker, 535 (69%) still contain the word "market", 105 "national" and 65 "EEA", i.e. many express scope in wording the marker list does not cover ("the market is national in scope"); 242 (31%) do not mention a market.

7.2. Source Fidelity

Of the 4,067 verifiable records (Table 3), 3,221 (79.2%) are verbatim, 349 (8.6%) near-verbatim, 26 (0.6%) partial and 471 (11.6%) absent. Absence is concentrated in documents that contain no market analysis: 136 of the 699 documents have at most three pages, and 353 of their 380 records are absent (the other 27 are copied boilerplate such as

party descriptions). Of the 118 absent records from longer documents, 26 are copies of the prompt's own examples (Section 8); 32 come from twelve documents of four to seven pages, eleven of which yielded no copied record at all; and 60 come from 19 longer documents: ancillary decisions that refer back to a main clearance decision, whose records restate the main decision's definitions in words that do not occur in the document, and a few long decisions whose records share only fragments with the text (coverage below 0.5). Of the 349 near-verbatim records, 292 have at most two missing windows, i.e. a single break; inspection of such cases in twelve documents found two causes, neither of them paraphrase: footnote text interleaved at a page break in the PyPDF2 text (as in the worked example), and a skipped intermediate paragraph (e.g. a party's submission between the Commission's precedent and its conclusion in M.10745).

Of the 48 reference records whose documents could be retrieved, 44 are verbatim, three near-verbatim and one absent: the record for M.10993, a three-page short-form decision, consists of plausible telecommunications market-definition language that does not occur in the decision. The manual check behind the reference set thus missed one fabricated record. The structural test does not detect fabrication: 336 of the 471 absent records pass it (71%), close to the rate for verbatim records (2,475 of 3,221, 77%); fabricated passages are, by construction, well-formed Commission prose.

7.3. Manual Inspection

Of the 50 records that pass the structural test, 49 were labelled A1 and one C (a competitive-assessment conclusion); one of the 49 is, however, a leaked prompt example (Section 8): face-valid and well-formed, but not from the decision it is attributed to. Of the 50 failing records, 38 were labelled A1 and three A2, i.e. 41 (82%; interval 69–90%) are market-definition content; one is a bare heading (B) and eight (16%) are not market definitions (C): remedies reasoning, a party description, Treaty text, and competitive assessment. Seven of the 41 are complete one-sentence scope statements below the 26-word minimum, such as "For AIs, the Commission therefore takes the view that the markets are at least EEA-wide"; four of them fail on length alone. One is a coherent 727-word party submission that fails only the maximum. The structural test's failures are thus mostly wording and length mismatches, as reviewers suspected, and its passes mostly genuine definitions; neither verdict is evidence about fidelity.

8. Error Analysis and Fixes

Short-form decisions that evade the filter. Most absent records, 353 of 471, come from simplified-procedure clearances. These contain the exclusion phrases, but PyPDF2 renders them with inserted spaces and line breaks ("on a simpli fied procedure", "Merger Regulation"), so exact matching fails: in a 40-document pilot sample, none of eight three-page documents matched a phrase, and the two-page document that did was not excluded because the rule applied only to three-page documents. The documents then reached call 1, whose prompt presupposes a market-definition section, and call 2, whose prompt says "you will see many market definitions". For a decision with no market analysis the model produced precedent-style passages that read like Commission text; eight records in the collection are the simplified-procedure recital itself. The released

code normalises whitespace before matching and applies the rule to documents of at most three pages, which excludes all nine short-form documents of the pilot; the fidelity script drops absent records and, optionally, every record from documents of at most three pages, which produced the cleaned release.

Prompt-example leakage. The four example passages in the extraction prompt were emitted as records for decisions that are not their source: 30 records across 11 cases reproduce example text verbatim. They pass the structural test and a face-validity reading; the fidelity test rejects all 30. The examples will be removed from the prompt; the effect on extraction quality is untested.

Boundaries, duplicates and wording. Near-verbatim records show the model splicing across footnotes and occasionally omitting a paragraph; length failures show the opposite boundary problem (261 records under 26 words, mostly single conclusion sentences and seven bare headings; 14 over 668 words); 117 exact duplicates mostly reflect one passage emitted under two topics. These are segmentation issues, not fidelity failures, for a gold-standard study to score. Most remaining structural failures are genuine definitions in wording the 18 markers do not cover, notably "national in scope"; extending the marker list would raise the rate without changing the collection, so we report the manual inspection instead.

9. Limitations and Future Work

No gold standard. Precision, recall and F1 against an independently annotated set of all definitions in a decision have not been measured: the fidelity test shows that 88% of records are copied from their source, not that they are the right passages or that none were missed. The sheet and protocol in `make_annotation_sheet.py` are ready for an expert annotation of ten decisions. *No baselines.* The two-stage design, the choice of Gemini 2.0 Flash (taken for its free-tier rate limits and long context) and the claim that rule-based extraction is brittle are stated, not demonstrated. A single-call variant and a heading-based rule baseline (`baselines/`) feed the same evaluation scripts, but neither they nor other LLMs have been run yet. *Single annotation.* The manual inspection and the reference set were each labelled once, without a second, independent annotator. *Scope.* Only English-language decisions of one authority were processed, and PDF text quality bounds output quality.

10. Conclusion

Lextract turns heterogeneous Commission decisions into 4,105 structured market-definition records with a two-call LLM pipeline whose prompts, settings and filters are now fully documented. The joint lesson of the structural test, the fidelity test and the manual reading is the one the workshop's call anticipates: the LLM copies the right kind of passage most of the time (79% verbatim, 9% near-verbatim across the collection), but it also fabricates fluent definitions when handed a decision that has none and leaks prompt examples into the output, and a resemblance-based check sees neither failure. A deterministic check against the source text does, at negligible cost, and it also caught a fabricated record in a manually verified reference set. We regard such a test as a necessary component of any LLM extraction pipeline over legal documents.

Acknowledgements. The author is grateful to Professor Thibault Schrepel of Stanford Law School for his advisement and guidance, and to the JURIX 2026 reviewers. This research received no funding from any government agency, university, company, or non-profit organization.

References

[1] European Commission. Commission Notice on the definition of the relevant market for the purposes of Union competition law. Official Journal of the European Union; 2024. OJ C, C/2024/1645, 22.2.2024.

[2] Bernhardt L, Dewenter R. The Impact of the More Economic Approach on EU Merger Decisions. Stanford Computational Antitrust. 2024. Available from: https://law.stanford.edu/wp-content/uploads/2024/12/Bernhardt-Dewenter.pdf.

[3] Affeldt P, Duso T, Szücs F. 25 Years of European Merger Control. International Journal of Industrial Organization. 2021;76:102720. Available from: https://doi.org/10.1016/j.ijindorg.2021.102720.

[4] Caselex. Caselex Market Definitions Module; 2026. Commercial product, https://www.caselex.eu/services/service.

[5] Agnoloni T, Bacci L, Peruginelli G, van Opijnen M, van den Oever J, Palmirani M, et al. Linking European Case Law: BO-ECLI Parser, an Open Framework for the Automatic Extraction of Legal Links. In: Legal Knowledge and Information Systems: JURIX 2017. vol. 302 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2017. p. 113-8. Available from: https://doi.org/10.3233/978-1-61499-838-9-113.

[6] Darji H, Mitrović J, Granitzer M. A Dataset of German Legal Reference Annotations. In: Proceedings of the Nineteenth International Conference on Artificial Intelligence and Law. ICAIL '23. New York, NY, USA: Association for Computing Machinery; 2023. p. 392-6. Available from: https://doi.org/10.1145/3594536.3595173.

[7] Janatian S, Westermann H, Tan J, Savelka J, Benyekhlef K. From Text to Structure: Using Large Language Models to Support the Development of Legal Expert Systems. In: Legal Knowledge and Information Systems: JURIX 2023. vol. 379 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2023. p. 167-76. Available from: https://doi.org/10.3233/FAIA230962.

[8] Costa MZ, Robson DF, Vieira TBP, Bourguet JR, Guizzardi G, Almeida JPA. Automated Semantic Annotation Pipeline for Brazilian Judicial Decisions. In: Legal Knowledge and Information Systems: JURIX 2024. vol. 395 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2024. p. 226-38. Available from: https://doi.org/10.3233/FAIA241248.

[9] Spada I, Nai R, Audrito D, Trifiletti VSM, Sulis E. An Expert-Validated LLM Framework for Transforming Legal Procurement Texts into Actionable Data. In: Legal Knowledge and Information Systems: JURIX 2025. vol. 416 of Frontiers in Artificial Intelligence and Applications. IOS Press; 2025. p. 356-63. Available from: https://doi.org/10.3233/FAIA251610.

[10] Nguyen PM, Nguyen HT, Zin MM, Satoh K. Data Augmented Pipeline for Legal Information Extraction and Reasoning. In: Proceedings of the Twentieth International Conference on Artificial Intelligence and Law. ICAIL '25. New York, NY, USA: Association for Computing Machinery; 2025. p. 481-2. Available from: https://doi.org/10.1145/3769126.3769200.

[11] Breton J, Billami MM, Chevalier M, Nguyen HT, Satoh K, Trojahn C, et al. Leveraging LLMs for legal terms extraction with limited annotated data. Artificial Intelligence and Law. 2025. Available from: https://doi.org/10.1007/s10506-025-09448-8.

[12] Castano S, Ferrara A, Furiosi E, Montanelli S, Picascia S, Riva D, et al. Enforcing legal information extraction through context-aware techniques: The ASKE approach. Computer Law & Security Review. 2024;52:105903. Available from: https://doi.org/10.1016/j.clsr.2023.105903.

[13] Kalai AT, Nachum O, Vempala SS, Zhang E. Why Language Models Hallucinate. arXiv; 2025. Available from: https://arxiv.org/abs/2509.04664.

[14] European Commission. Commission Notice on a simplified procedure for treatment of certain concentrations under Council Regulation (EC) No 139/2004. Official Journal of the European Union; 2013. OJ C 366, 14.12.2013, p. 5.